Section 2.4 — Factoring: Trinomials and Special Forms — Answer Key

Math 103: College Algebra

Factoring Quadratic Trinomials

Each quadratic trinomial below can be written as a product of two linear binomials. Factor each quadratic trinomial.

1. $(3x+5)(x+1)$
2. $(2x+3)(x+3)$
3. $(2x-5)(x+2)$
4. $(3x-2)(2x+3)$
5. $(x+5)(x+7)$
6. $(x+7)(x-2)$
7. $(x-10)(x+2)$
8. $(x+6)(x+2)$
9. $(x+8)(x-5)$
10. $(x-7)(x-4)$
11. $(2x+1)(2x+5)$
12. $(5x+1)(4x+1)$
13. $(5x-9)(x-4)$
14. $(4x+5)(5x-16)$
15. $(5x-6)(x-3)$
16. $(2x-5)(x+10)$

Possibly Prime Quadratic Trinomials

Factor each quadratic trinomial, if possible. If it is prime, write "prime."

17. $(x-2)(x-3)$
18. Prime
19. $(x+4)(x-2)$
20. Prime
21. Prime
22. $(8x-5)(x+8)$
23. Prime
24. $(5x-1)(4x+3)$

Possible Coefficients

Using the AC method, list all nonnegative integer coefficients that can fill in the blank to allow the trinomial to be factored.

25. $9, 12, 21$
26. $1, 6, 14, 21, 34, 71$
27. 6
28. $21, 23, 33, 47, 91$
29. $3, 18, 42, 63, 73, 102, 158, 213, 322, 647$
30. $12, 18, 33$
31. $30, 33, 35, 42, 58, 75, 100, 217$
32. $8, 16$

Perfect Squares

For each trinomial below:

a. Determine whether two of its terms are perfect squares.

b. If so, determine whether the middle term makes it a perfect saure.

c. If it is a perfect square, factor it as such. (Don't factor if it isn't a perfect square.)

33. a. $x^2 = (x)^2$, $36y^2 = (6y)^2$ b. $2(x)(6y) = 12xy$ c. $(x+6y)^2$
34. a. $25x^2 = (5x)^2$, $100 = 10^2$ b. $2(5x)(10) \neq 12x$ c. Not square
35. a. Not square b. N/A c. N/A
36. a. $25z^2 = (5z)^2$, $36 = 6^2$ b. $-2(5z)(6) = -60z$ c. $(5z-6)^2$
37. a. $9q^2 = (3q)^2$, $4 = 2^2$ b. $2(3q)(2) \neq 38q$ c. Not square
38. a. $w^6v^2 = (w^3v)^2$, $25w^2v^6 = (5wv^3)^2$ b. $2(w^3v)(5wv^3) = 10w^4v^4$ c. $(w^3v + wv^3)^2$
39. a. $25 = 5^2$, $64x^2 = (8x)^2$ b. $-2(5)(8x) = -40x$ c. $(5-8x)^2$
40. a. $25x^2 = (5x)^2$, $25 = 5^2$ b. $2(5x)(5) \neq 42x$ c. Not square

Binomial Special Forms

For each binomial below:

a. Decide which, if any, binomial special form it is.

b. If it is, factor it.

41.	a. Difference of Squares	b. $(x-4)(x+4)$
42.	a. Not a special form	b. Not a special form
43.	a. Difference of Cubes	b. $(x-3)(x^2+3x+9)$
44.	a. Sum of Cubes	b. $(x+6)(x^2-6x+36)$
45.	a. Sum of Cubes	b. $(4x+1)(16x^2+4x+1)$
46.	a. Difference of Cubes	b. $(5-2y)(25+10y+4y^2)$
47.	a. Difference of Squares	b. $(2x-3u)(2x+3u)$
48.	a. Sum of Cubes	b. $(2+v^2)(4+2v^2+v^4)$
49.	a. Difference of Cubes	b. $(3-u^2)(9+3u^2+u^4)$
50.	a. Difference of Squares	b. $(7-uv^2)(7+uv^2)$
51.	a. Difference of Squares	b. $(x^2-6v^2)(x^2+6v^2)$
52.	a. Sum of Cubes	b. $(3z^2+v)(9z^4-3z^2v+v^2)$
53.	a. Sum of Squares (prime)	b. Prime
54.	a. Sum of Cubes	b. $(3z^2+4v)(9z^4-12z^2v+16v^2)$
55.	a. Difference of Squares	b. $(3x-6)(3x+6)$
56.	a. Sum of Squares (prime)	b. Prime
57.	a. Difference of Cubes	b. $(2z^2-5)(4z^4+10z^2+25)$
58.	a. Sum of Cubes	b. $(5+w)(25-5w+w^2)$